

Addition and subtraction of radicals



1

a $26\sqrt{3}$

b $-18\sqrt{5}$

c $27\sqrt{6} + 11\sqrt{13}$

d $20\sqrt{6} + 5\sqrt{11}$

e $4\sqrt{6}$

f $16\sqrt{5}$

g $-3\sqrt{2}$

h $42\sqrt{5}$

i $11\sqrt{10}$

j $-15\sqrt{5}$

k $28\sqrt{2}$

l $41\sqrt{3} + 12\sqrt{7}$

m $10\sqrt{3}$

n $-8\sqrt{3}$

o $4 + 3\sqrt{5}$

2

a $3\sqrt{3}$

b $3\sqrt{3}$

c $5\sqrt{2}$

d $6\sqrt{5}$

3

a $3\sqrt[3]{v}$

b $7\sqrt[3]{u}$

c $2x^2\sqrt{ax}$

Multiplication and division of radicals

4

a $\sqrt{22}$

b $4\sqrt{2}$

c $20\sqrt{7}$

d $15\sqrt{35}$

e $\sqrt{70}$

f $35\sqrt{21}$

g $\sqrt{35}$

h $80\sqrt{5}$

i $\sqrt{231}$

j $11\sqrt{5}$

k $20\sqrt{11}$

l $30\sqrt{55}$

m $14\sqrt{11}$

n $24\sqrt{15}$

o $320\sqrt{3}$

p $40\sqrt{51}$

q $340\sqrt{7}$

r $216\sqrt{17}$

5

a $\sqrt{5}$

b $2\sqrt{7}$

c 32

d $8\sqrt{11}$

e $4\sqrt{7}$

f $\sqrt{3}$

g $\sqrt{11}$

h $\sqrt{3}$

i $\sqrt{7}$

j $\sqrt{13}$

k $5\sqrt{7}$

l $10\sqrt{5}$

m $15\sqrt{2}$

n $2\sqrt{7}$

o 3

p 6

q 10

r 16

s 10

t $\frac{5}{9}$

u $\frac{9}{2}$

6

a 2

b $\frac{1}{\sqrt{5}}$

c 4

d $\frac{1}{\sqrt{3}}$

e $\frac{1}{\sqrt{3}}$

f 2

g $\frac{2}{3}$

h $\frac{3}{2}$

i $17\sqrt{7}$

7

a $4j\sqrt{k}$



b y^3

8

a $\frac{1}{7}$

b $\sqrt{\frac{p}{q}}$

c $3\sqrt{7}$

Mixed operations with radicals

9

a $-5\sqrt{7} + \sqrt{35}$

c $-42\sqrt{22} + 7\sqrt{77}$

e $3\sqrt{7} + \sqrt{21}$

g $3\sqrt{39} - 15\sqrt{3}$

i $-4\sqrt{77} + 4\sqrt{14}$

k $-24\sqrt{14} + 8\sqrt{6}$

m $63\sqrt{5}$

o $24\sqrt{77} - 32\sqrt{55}$

q $-24\sqrt{55} - 56\sqrt{77}$

b $7\sqrt{14} + 42\sqrt{2}$

d $\sqrt{77} + 4\sqrt{11}$

f $-6\sqrt{2} + \sqrt{22}$

h $\sqrt{33} + \sqrt{39}$

j $15\sqrt{11} + 3\sqrt{55}$

l $15\sqrt{10} + 20\sqrt{14}$

n $11\sqrt{15}$

p $\sqrt{143} + 2\sqrt{13}$

10

a $\sqrt{26} - 3\sqrt{13} + 5\sqrt{2} - 15$

c $12 + 2\sqrt{11}$

e $40 - 5\sqrt{3} - 8\sqrt{7} + \sqrt{21}$

g -110

i 25

b $\sqrt{14} - \sqrt{77} - \sqrt{26} + \sqrt{143}$

d $29 + 12\sqrt{5}$

f $104 - 6\sqrt{55}$

h 3

11

a 60

b $\frac{12\sqrt{3}}{\sqrt{5}}$

c $\sqrt{5} + \sqrt{7}$

d $\frac{3\sqrt{3}}{\sqrt{10}}$

12

a $x = 5999, y = -560$

b $x = 5, y = -3$

13

a $42 + (8 + 2m)\sqrt{13} + m^2$

b $m = -4$

14

a u

b $3u$



c $10\sqrt{u}$

d $6\sqrt{u} + \sqrt{v}$

e $15\sqrt{a}$

f $3\sqrt{5a}$

Rationalize the denominator

15

a No

b Yes

c No

16

a $\frac{\sqrt{11}}{11}$

b $\frac{\sqrt{22} - 2\sqrt{11}}{11}$

c $\frac{7\sqrt{12}}{12}$

d $\frac{14\sqrt{143}}{117}$

e $-\frac{3\sqrt{13}}{65}$

f $7\sqrt{3}$

g $\frac{10\sqrt{35} - \sqrt{5}}{5}$

h $\frac{\sqrt{3}}{3}$

17

a $2 - \sqrt{7}$

b $8 + \sqrt{u}$

c $4 + 3\sqrt{10}$

d $-5\sqrt{3} - 4\sqrt{v}$

18

a $\frac{-9(\sqrt{5} + 7)}{44}$

b $4(\sqrt{10} + \sqrt{3})$

c $\frac{5\sqrt{42} - 4\sqrt{14}}{118}$

d $\frac{13 + 2\sqrt{22}}{9}$

e $\frac{887 + 144\sqrt{35}}{247}$

f $\frac{4(9\sqrt{11} + 5)}{433}$

g $\frac{9 + 2\sqrt{14}}{5}$

h $\frac{\sqrt{10} - 1}{9}$

19

a $\frac{6\sqrt{10} + 5}{20}$

b $\frac{3\sqrt{5} - 15\sqrt{2}}{20}$

c $-\frac{11\sqrt{15}}{18}$

d $-\frac{2\sqrt{2}}{23}$

20 No solution provided

21

a $\frac{\sqrt{3} - 1}{2}$

b 16

22

a $19 + 8\sqrt{3}$

b $\frac{6 - \sqrt{3}}{3}$

23 $\sqrt{1800}$

Applications



24 $12\sqrt{2}$ units

25 $3\sqrt{5}$ units

26

a $4\sqrt{7}$ ft

b 3 ft^2

27 $c = \sqrt{26}$ units

28 $9\sqrt{3}$ cm

29

a $\frac{\sqrt{5} + 1}{2}$

b 1.618

30 $4\sqrt{10} - 12$ m

31

a 17

b $P = \frac{12\sqrt{x} + x}{5}$

c 17

32

a $\frac{\sqrt{3Vh}}{h}$

b 4 in